BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Other Cell Types

Species Used *Leishmania,* all species within the genus

Molecules Electroporated DNA: supercoiled & linear; 3.3 - 33 kb pY, PR-neo.

Before the Pulse

Cell Growth Medium M199 (see Kapler *et. al.* 1990)

Growth Phase at Harvest Late log

Pre-pulse Incubation up to 2 hours

Wash Solution 21mM HEPES,pH 7.5,0.7mM Na2PO4, 137mM NaCl, 6 mM glucose,5mM KCl

The Pulse

Instruments Used Not given

Electroporation Temperature Cells, cuvettes, DNA on ice; 0 °C

Electroporation Medium 21 mM HEPES, pH 7.5, 0.7 mM Na2PO4, 137 mM NaCl, 6 mM glucose, 5 mM KCl

Cell Density 10 (8) / ml

Volume of Cells 0.4 ml

DNA Concentration 300 to 1000 µg / ml

DNA Resuspension Buffer TE

Volume of DNA 1 to 100 µg

Cuvette Gap 0.2 cm

Voltage 0.45 kV

Field Strength 2.25 kV/cm

Capacitor 500 µF

Resistor (Pulse Controller) none Ω

Time Constant ~ 4 msec

After the Pulse

Outgrowth Medium M199 medium

Outgrowth Temperature 26 °C

Length of Incubation Overnight

Selection Method or Assay Used G418, hygromycin, gancyclovir β-glactosidase, β-glucuronidase

Electroporation Efficiency 10 to 60 transformants / µg DNA, up to10 (-4) / cell

Per Cent Survival 50%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Kapler, *et. al.*, 1990 *Molec. Biol.* **10**:1087 (G418).
Cruz & Beverly, 1990 *Nature* **348**:171. Gene replacement.
LeBowitz, *et. al.*, 1990 *PNAS* **87**: 9736. Expression vector.
Coburn, *et. al*., 1991*Molec. Bioch. Parasitology* **46**:169 (diverse species).
LeBowitz, *et. al.*, 1991 *Gene***103**:119-123. b-gal, b-gluc reporters.
Cruz, *et. al*., 1991 *PNAS* **88**:7170-7174. Hygromycin & gene replacement.

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